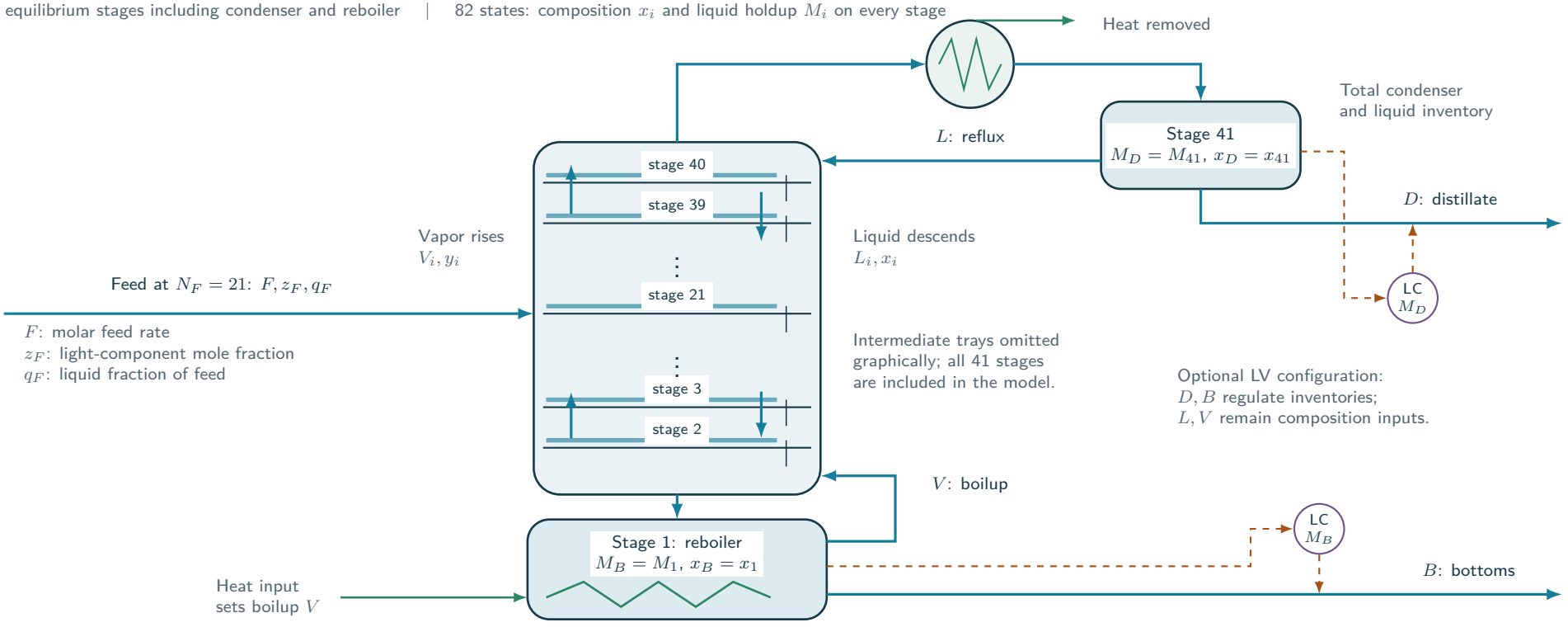
Column A: binary distillation

Paper section 3.10: *Continuously operated binary distillation column*

41 equilibrium stages including condenser and reboiler | 82 states: composition x_i and liquid holdup M_i on every stage



Interior-stage conservation

$$\dot{M}_i = L_{i+1} - L_i + V_{i-1} - V_i$$
$$\frac{d(M_i x_i)}{dt} = L_{i+1}x_{i+1} + V_{i-1}y_{i-1} - L_i x_i - V_i y_i.$$

Feed, condenser and reboiler use separate balances.

Thermodynamics and hydraulics

$y_i = \alpha x_i / [1 + (\alpha - 1)x_i]$, $\alpha = 1.5$.
$L_i = L_{0i} + (M_i - M_{0i})/\tau_l$ on trays.

Constant pressure; negligible vapor holdup; constant molar flows in each section.

Interfaces and units

4×4: inputs L, V, D, B; outputs x_D, x_B, M_D, M_B.
Flows: kmol/min; holdups: kmol.

NTNU nominal values: $L_0 = 2.70629$, $V_0 = 3.20629$; $\tau_l = 0.063$ min.

Scope: Column A, Eqs. (47)–(54), aligned with Skogestad's NTNU reference code. Incoming vapor uses V_{i-1}, correcting the printed Eq. (47). Dashed D/B level loops describe the documented LV extension.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.